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ELAPSE

Entropy-Linked Autonomous Propagation and Self-Erasure

A smarter way to make sensitive data delete itself
before it spreads too far

Mathematical Modelling Phase 1 EL Presentation

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Once you share a secret... can you really take it back?



Data escapes

The moment a file is opened or forwarded, you lose control. It can copy itself across hundreds of devices in seconds.



Timers are blind

Existing tools delete data after a fixed time (like 8 hours) — but by then, it may have spread everywhere already.



Networks are unpredictable

Every network (company, social, P2P) spreads data differently. A one-size-fits-all rule fails most of the time.

Bottom line: we need a system that watches how data spreads in real time — and acts before it's too late.

What others have tried

Research Area	Key Work	What's Missing
Self-destructing data	Vanish, FADE, Ephemerizer	All use fixed timers — ignore how fast data actually spreads
Epidemic models	SIR / SIS models on networks	Great for describing spread , but not designed for deletion
Shannon entropy on graphs	Zhao et al. 2024, Shuman et al. 2013	Measures spread well — but no one used it to trigger deletion
Machine unlearning	Bourtoule 2021, Juliussen 2023	Focuses on AI models , not live network data propagation
Graph diffusion (PDE)	GRAND 2021, Eliasof 2024	Models spread beautifully — but not applied to active containment

EXISTING SYSTEMS & THEIR LIMITS



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Vanish (2009)	FADE (2010)	Ephemerizer (2005)	Machine Unlearning
Hides decryption key over time in DHT	Cloud storage with guaranteed deletion	Data expires after set timeout	Removes data from AI model memory
⚠ Broken by Sybil attacks	⚠ Still uses fixed timers	⚠ Blind to actual spread	⚠ Not for live networks

🔑 The shared flaw in ALL existing systems: they decide when to delete based on the clock — not on how widely the data has actually spread.

ELAPSE fixes this by watching the spread pattern itself, not the clock.

THE BIG IDEA

Innovation & Novelty

Think of it like a
forest fire alarm 🌲🔥

Old way: alarm goes off after 8 hours no matter what.

ELAPSE way: alarm goes off the moment the fire has spread to too many trees — regardless of how long it took.



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Entropy as a ruler

We measure how spread-out the data is using Shannon Entropy — a mathematical score. High score = data is everywhere = delete now.

5 deletion engines in 1

We borrow deletion strategies from physics, biology, finance, epidemiology, and sociology — and blend the best of all five.

Self-tuning weights

The system automatically learns which strategy works best for each type of network. No manual tuning needed.

HOW ELAPSE WORKS - METHODOLOGY



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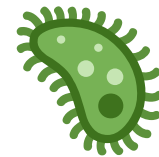
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M1 Diffusion



Tracks how data flows through the network like water through pipes

M2 Epidemic



Models data spread like a virus — and triggers deletion like a cure

M3 Finance



Treats data risk like a stock price about to crash through a floor

M4 Biology



Uses cooperative binding (like cells) for a smooth, stable deletion trigger

M5 Social



Deletes data when enough neighbours in the network have already deleted it



M6 ELAPSE Ensemble — blends all 5 engines with smart auto-tuned weights

OBJECTIVES



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01

Replace the clock with topology

Instead of deleting after X hours, delete when the data has spread to too many places — measured using Shannon entropy.

02

Work across different networks

Make sure the system works whether the network is random, hub-based (like social media), or small-world (like company networks).

03

Prove it mathematically

Show formally that combining multiple deletion strategies always performs at least as well as any single one on its own.

04

Auto-tune for each network

Use optimization to automatically find the best mix of deletion engines for any given network — no human tuning.

05

Outperform existing protocols

Beat time-based protocols like Vanish by containing data spread 7.5× better, validated across 500-node network simulations.

RESULTS AT A GLANCE



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93.3%

Less privacy exposure
(vs. no deletion)

7.5×

Better than Vanish-style
timed protocols

3 types

of networks tested
(random, hub, small-
world)

500

nodes — largest
network tested

IEE (Time-Integrated Entropy Exposure) Score (lower = better privacy protection)

No deletion (M0)

7,055

Vanish-style timer

3,528

Best single engine (M4)

532

ELAPSE Ensemble (M6)

470

$$\text{IEE}(\mathbf{w}) = \int_0^T H(\mathbf{p}(t; \mathbf{w})) \cdot M(t; \mathbf{w}) dt \quad (9)$$

where $M(t; \mathbf{w}) = \sum_{i=1}^n x_i(t)$ represents the aggregate surviving data mass remaining in the network at time t .

PAPER ABSTRACT



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The restriction of information flow across stochastic, decentralized networks presents a significant challenge for ongoing data privacy. Current cryptographic access controls are frequently bypassed post-exfiltration, and static active containment protocols generally rely on chronological decay sequences that do not account for the topological realities of viral spread. In this study, we introduce the ELAPSE framework, shifting from chronological data lifetimes toward spatial-topological containment. By measuring the Shannon entropy of the data's spatial dispersion, ELAPSE links the mortality rate directly to structural privacy loss. To address the failure of single domain deletion triggers across heterogeneous network spectra, we construct an optimization-driven ensemble that unifies five distinct physical, epidemiological, financial, biological, and sociological switching mechanisms. The ensemble weights are continuously resolved via an entropy-regularised Nelder-Mead optimization to minimise the Time-Integrated Entropy Exposure (IEE). We prove an Ensemble Dominance Proposition demonstrating that the unregularised system mathematically bounds the best baseline deletion rate, and introduce a regularisation parameter to ensure weight diversity. Numerical simulations spanning Erdős-Rényi, Barabási Albert, and Watts-Strogatz networks for $n \in [50, 500]$ confirm the framework's efficacy. While uncontained diffusion (the theoretical baseline) resulted in exponential privacy exposure (IEE = 7055.7 at $n = 500$), the regularised ELAPSE ensemble bounded total exposure to IEE = 470.4, confirming its viability for decentralized data architecture.

CONCLUSION



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ELAPSE shows that data can be made to delete itself at the right time — not just after a fixed delay, but when it has actually spread too far.

- ✓ Replaces clocks with real-time spread measurement
- ✓ Blends 5 domain-inspired deletion strategies into one smart ensemble
- ✓ Mathematically proven to beat any single strategy on its own
- ✓ 93.3% less privacy exposure vs. uncontrolled spread

Future work: real-world deployment on live networks, neural weight prediction, directed graph support